Experiment No. 1 – Introduction to Linux Commands

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Aim

To study and practice basic Linux commands for file and directory operations, user management, and permission management.

Objectives

- Understand the Linux operating system environment.
- Perform file and directory operations using Linux commands.
- Create and manage users and groups.
- Understand and modify file permissions.

Theory

Linux is an open-source operating system widely used in servers, cloud computing, and DevOps environments. Linux provides a Command Line Interface (CLI) through which users can perform various administrative and development tasks efficiently.

Linux commands are categorized into:

1. File and Directory Management Commands
2. User Management Commands
3. Permission Management Commands
4. System Information Commands

Part A: File and Directory Operations

1. Display Current Directory

pwd

Output:

/home/student

2. List Files and Directories

ls

Detailed listing:

ls -l

Show hidden files: **ls -la**

3. Create Directory

```
mkdir DevOpsLab
```

4. Change Directory

```
cd DevOpsLab
```

5. Create File

```
touch sample.txt
```

6. Display File Content

```
cat sample.txt
```

7. Copy File

```
cp sample.txt backup.txt
```

8. Rename File

```
mv sample.txt file1.txt
```

9. Delete File

```
rm file1.txt
```

10. Delete Directory

```
rmdir TestFolder
```

Part B: User Management

1. Display Current User

```
whoami
```

2. Create New User

```
sudo useradd student1
```

3. Set Password

```
sudo passwd student1
```

4. View User Information

```
id student1
```

5. Switch User

```
su student1
```

6. Delete User

```
sudo userdel student1
```

Part C: Permission Management

1. View File Permissions

ls -l

Example Output:

-rw-r--r-- 1 user user 100 Jan 1 10:00 sample.txt

Permission Meaning

Symbol	Permission
r	Read
w	Write
x	Execute

Permission Categories

Category	Meaning
User (u)	Owner
Group (g)	Group Members
Others (o)	All Users

2. Change Permissions

Grant execute permission:

chmod +x sample.sh

Set permission using numeric mode:

chmod 755 sample.sh

3. Change File Ownership

sudo chown student1 sample.txt

Sample Commands Summary

Command	Purpose
pwd	Show current directory
ls	List files
mkdir	Create directory
cd	Change directory
touch	Create file
cat	Display file content
cp	Copy file
mv	Move/Rename file
rm	Delete file
whoami	Display current user
useradd	Add user
passwd	Set password
chmod	Change permissions
chown	Change ownership